

Chapter 1 Algebraic Groups

1.1 Definitions

1.1.1 Group Structure

Definition 1.1.1.1 An **algebraic group** is a variety G with group struction

$$m : G \times G \rightarrow G$$

$$e : * \rightarrow G$$

$$i : G \rightarrow G$$

And:

- m, e, i are regular;
- The following diagrams commutes:

$$\begin{array}{ccccc} G \times G \times G & \xrightarrow{id \times m} & G \times G & * \times G & \xrightarrow{e \times id} & G \times G & \xleftarrow{id \times e} & G \times * & G & \longrightarrow & G \times G & \xleftarrow{id \times ie} & G \\ \downarrow m \times id & & \downarrow m & & \searrow \simeq & \downarrow m & & \swarrow \simeq & \downarrow e \times id & & \downarrow m & & \downarrow \\ G \times G & \xrightarrow{m} & G & & & G & & & * & \xrightarrow{e} & G & \xleftarrow{e} & * \end{array}$$

Here: $*$ is the one-point vareity $Spec(k)$.

Similarly, an **algebraic monoid** over k is an algebraic scheme M with m, e commutes.

If G is affine as a scheme, then G is called **linear(affine) algebraic group**.

Definition 1.1.1.2

- Let $(G, m_G), (F, m_F)$ be two algebraic groups. We say that G is an F -group if G is a F -scheme.
- Morphsims m, inv are defined over F ;
- Thus, the F -homomorphisms and F -groups are well-defined.

Definition 1.1.1.3 If G is an F -group. Set

$$G(F) := Hom_{Spec(F)}(Spec(F), G)$$

which has a canonical group structure.

♣ **Example 1.1.1.4** Let G be a linear algebraic group. Then put $A = k[T]$ The homomorphism

$$\Delta : k[T] \rightarrow k[T] \otimes_k k[T] \simeq k[T, U] \quad \iota : k[T] \rightarrow k[T] \quad e : k[T] \rightarrow k$$

is given by

$$\Delta T := T + U$$

$$\iota T := -T$$

$$eT := 0$$

Denote this algebraic group as G_a . It is the additive group, and $F[T]$ defines an F -structure on G_a (consider the scheme morphism induced by those ring maps), for any subfield F of k . (The morphisms are F -morphisms)

♣ **Example 1.1.1.5** (Laurent Series) Define: $G = A^1 - \{0\} = k^*$, with multiplication as group operation. Now define

$$A = k[T, T^{-1}]$$

and

$$\begin{aligned} \Delta : k[T, T^{-1}] &\rightarrow k[T, T^{-1}] \otimes k[T, T^{-1}] = k[T, T^{-1}, U, U^{-1}] & \Delta T &:= TU \\ \iota : k[T, T^{-1}] &\rightarrow k[T, T^{-1}] & \iota T &:= T^{-1} \end{aligned}$$

Denote G_m or GL_1 . With F -structure by $F[T, T^{-1}]$.

Especially: Let $n \in \mathbb{Z}^\times$, there exists a homomorphism of algebraic groups

$$\begin{aligned} \phi : A &\rightarrow A \\ x &\mapsto x^n \end{aligned}$$

Induces a homomorphism of schemes $(\phi, \phi^\#) : G_m \rightarrow G_m$.

♣ **Example 1.1.1.6** (Matrices Monoid) Define the matrices $M_n := k^{n^2}$. Then, the determinant function is well-defined.

♣ **Example 1.1.1.7** (General Linear Groups) Then, define:

$$GL_n := \{X \in M_n \mid D(X) \neq 0\}$$

with matrix multiplication as group operation.

♣ **Example 1.1.1.8** (Special Linear Group) Define

$$SL_n := \text{Spec}(k[T_{1,1}, \dots, T_{n,n}]/(\det(T_{ij}) - 1))$$

with matrix multiplication

♣ **Example 1.1.1.9** (Orthonormal Groups)

♣ **Example 1.1.1.10** (Special Orthonormal Groups)

♣ **Example 1.1.1.11** (Special Symplectic Groups)

♣ **Example 1.1.1.12** (Upper Triangular Groups)

♣ **Example 1.1.1.13** (Lower)

♣ **Example 1.1.1.14** (Unipotent Upper Triangular Matrices)

♣ **Example 1.1.1.15** **Nonlinear Algebraic Groups: Elliptic Curves** Consider the curve in projective plane: $V(F) \subseteq \mathbb{P}^2$, where

$$F = X_0 X_2^2 - X_1^3 - a X_1 X_0^2 - b X_0^3$$

which stands for the polynomial

$$f = T^3 + aT + b$$

has no multiple roots.

Group Operations:

Homomorphisms

Definition 1.1.1.16 A homomorphism $\varphi : (G, m) \rightarrow (G', m')$

Subgroups

Definition 1.1.1.17 An **algebraic subgroup** of an algebraic group (G, m_G) over k is an algebraic group

$$(H, m_H)$$

over k such that:

- H is a k -subscheme of G ;

- The inclusion map is a homomorphism of algebraic groups.

Homogeneity

Notation For an algebraic scheme X over k , we write $|X|$ for the underlying topological space. And

$$\kappa(x) := \mathcal{O}_{X,x}/m_x, x \in |X|$$

Then, define:

$$X(k) := \{x \in |X| \mid \kappa(x) = k\}, k \in K$$

Now, let (G, m) be an algebraic group over k , define:

$$G(k) := \text{Hom}_{\text{Sch}/k}(\text{Spec}(k), G)$$

with $m(k) : G(k) \times G(k) \rightarrow G(k) \rightarrow G(k)$ makes $G(k)$ into a group with neutral element $e(*)$ and inverse map $\text{inv}(k)$. (Constructions: Consider $x, y \in G(k) : \text{Spec}(k) \rightarrow G$. By the universal property of product

$$(x, y) : \text{Spec}(k) \rightarrow G \times_k G$$

Then, the product

$$\text{Spec}(k) \xrightarrow{(x,y)} G \times_k G \xrightarrow{m} G$$

And identity:

$$e : \text{Spec}(k) \rightarrow G$$

Inverse Map: Suppose $(x : \text{Spec}(k) \rightarrow G) \in G(k)$, we have

$$\text{Spec}(k) \xrightarrow{x} G \xrightarrow{\text{inv}} G$$

)

♠ **Proposition 1.1.1.18** Especially, if k is algebraically closed we have

$$G(k) \cong G$$

◦ **Proof ...**

□

Commutativity

Definition 1.1.1.19 An algebraic group (G, m) is **commutative** if

$$m \circ t = m$$

where t is the transposition map $(x, y) \mapsto (y, x)$.

...

◦ **Proof ...**

□

Group Actions

Definition 1.1.1.20 Let (G, m) be an algebraic group over k . For each $a \in G(k)$, there is a translation map

$$l_a : G \simeq \{a\} \times G \xrightarrow{m} G$$

Algebraic Groups over Rings

Definition 1.1.1.21 Let R be a (finitely generated) k -algebra. Formally, an **algebraic scheme** over R is a scheme X equipped with a morphism

$$X \rightarrow \operatorname{Spec}(R)$$

of finite type.

Less formally, we can think of X as an algebraic scheme over k such that $\mathcal{O}_X$ is equipped with an R -algebra structure compatible with its k -algebra structure.

Definition 1.1.1.22

Normal and Characteristic Subgroups

Definition 1.1.1.23 ...

Normalizers

Centralizers

Complements

Kernels

Definition 1.1.1.24 Let $\varphi : G \rightarrow H$ be a homomorphism of algebraic groups. And let

$$\ker(\varphi) = G \times_H *$$

$$G \quad H$$

Group Actions

Closed Subfunctors

1.1.2 Scheme Theoretic Properties

Density of Points

Review:

Definition 1.1.2.1

Smoothness

♠ **Proposition 1.1.2.2** The following conditions on an algebraic group G are equivalent:

- G is smooth;
- G° is smooth;
- The local ring $\mathcal{O}_{G,e}$ is regular;

- The tangent space $T_e(G)$ to G at e has dimension $\dim(G)$;
- G is geometrically reduced;
- for all k -algebra R and all ideals I in R such that $I^2 = 0$. The map

$$G(R) \rightarrow G(R/I)$$

is surjective.

◦ **Proof ...**

□

★ **Corollary 1.1.2.3**

- For an algebraic group G ,

$$\dim(G) \leq \dim T_e(G)$$

- Equal $\Leftrightarrow G$ is smooth.

◦ **Proof ...**

□

Algebraic Group vs Abelian Variety

Rigidity Lemma

Constructible Sets, Closure

1.1.3 Properties of Algebraic Groups

Uniqueness

♠ **Proposition 1.1.3.1** The map $e :$

◦ **Proof ...**

□

Separatedness

♠ **Proposition 1.1.3.2** (Separatedness)

◦ **Proof ...**

□

Identity Component

Recall: An algebraic scheme over a field is a finite disjoint union of its (closed-open) connected components.

Definition 1.1.3.3 For an algebraic group G , let G° of G is an algebraic subgroup of G , which contains $e \in G$. Then, it is called **identity/neutral** component of G .

♠ **Proposition 1.1.3.4** ...

◦ **Proof ...**

□

Fundamental Group

◇ **Lemma 1.1.3.5** Let X be a connected algebraic scheme over k such that

$$X(k) \neq \emptyset$$

Then, X is geometrically connected. Moreover, for any algebraic scheme Y over k ,

$$\pi_0(X \times Y) \simeq \pi_0(Y)$$

In particular: $X \times Y$ is connected if Y is connected.

◦ **Proof ...**

□

★ **Corollary 1.1.3.6** A connected algebraic group is irreducible.

◦ **Proof ...**

□

♠ **Proposition 1.1.3.7** Summary: The following conditions on an algebraic group G over k are equivalence:

-
-
-
-

Chevalley's Theorem on Algebraic Groups

Identity Components

Closed Subgroup of Finite Index

1.1.4 Examples and Constructions

Affine Algebraic Groups

Definition 1.1.4.1

- (Additive Group) The additive group G_a is the functor

$$R(R, +)$$

It is represented by

- (Multiplicative Group) The multiplicative group G_m is the functor

$$R(R^\times, \cdot)$$

Especially: Let F be a finite group. The **text**

General Linear Group

Definition 1.1.4.2 The **general linear group** GL_n is the functor

$$R \mapsto GL_n(R)$$

($GL_n(R)$ is the multiplicative group of invertible $n \times n$ matrices) It is represented by:

$$\mathcal{O}(GL_n) := k[T_{1,1}, \dots, T_{1,n}, T_{2,1}, \dots, T_{nn}, T] / (\det(T_{ij})T - 1) = k[T_{1,1}, \dots, T_{n,n},]$$

and: The universal element

1.2 Affine Algebraic Groups, and Hopf Algebras

1.2.1 The Comultiplication Map

1.3 Quotient

1.4 title

1.5 Lie Algebra of Algebraic Group

Chapter 2 Finite Group Scheme